



DAYANANDA SAGAR COLLEGE OF ENGINEERING, BANGALORE

Department of Electronics and Telecommunication Engineering

IV Sem Mini Project (22ET49) AY- 2025-26

Autonomous RL for Resonator Calibration

**A Quantum-Inspired
Control System**

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1DS24ET001
1DS24ET008
1DS24ET012
1DS24ET050



THE THREAT

Resonant transition frequencies of qubits continuously drift due to thermal and magnetic noise.

THE BOTTLENECK

Brute-force spectroscopic sweeps are computationally intractable for hundreds of qubits.

THE MANDATE

Sustained coherent operation requires autonomous, closed-loop tracking without periodic downtime.

Bridging the Gap: The Lorentzian Universality Hypothesis

Quantum Reality

Component:
Superconducting Transmon Qubit

Mechanism:
Fluctuating Josephson Junction

Noise Source:
Stochastic Flux Noise (4–8 GHz)

$$A(f) = \frac{1}{1 + Q^2 \left(\frac{f - f_0}{f_0} \right)^2}$$

Classical Analogue

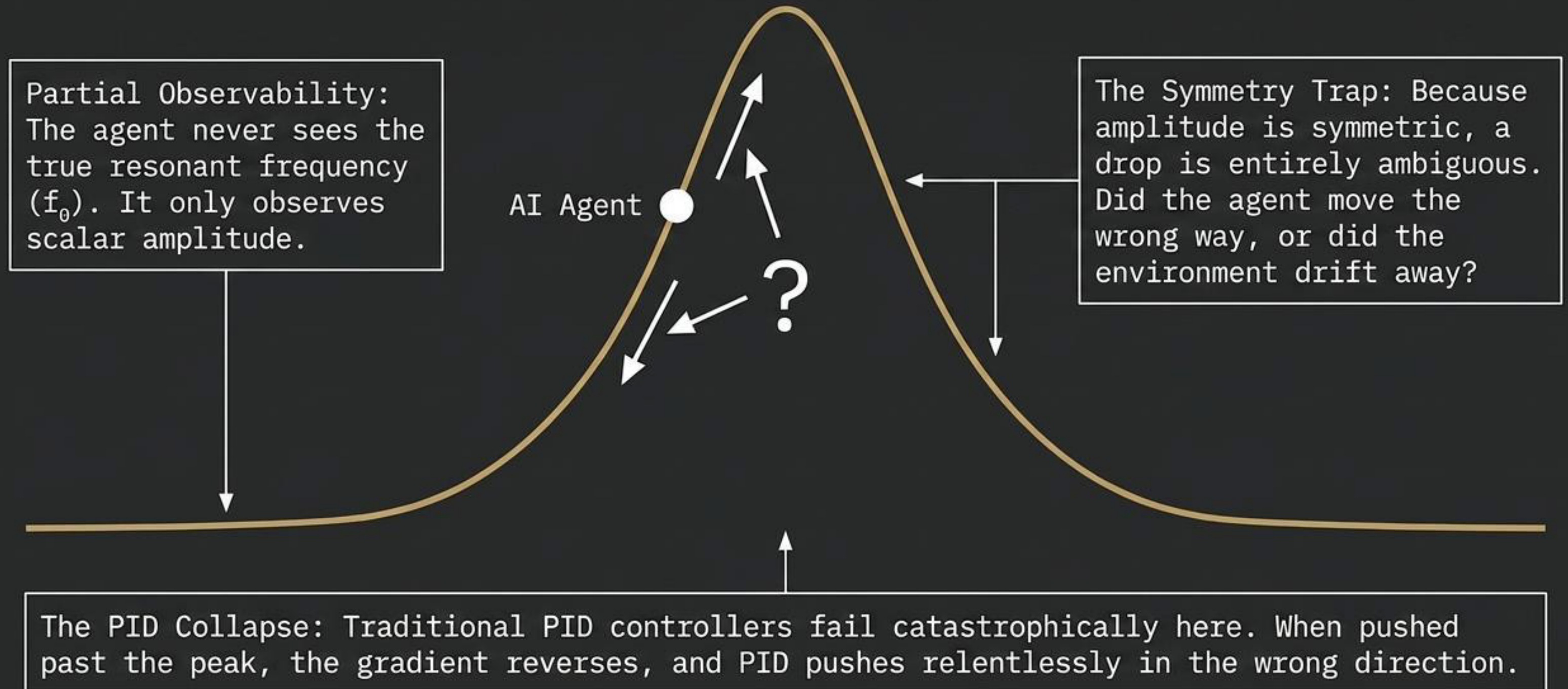
Component:
LC Tank Resonator

Mechanism:
BB909/MV209 Varactor Diode

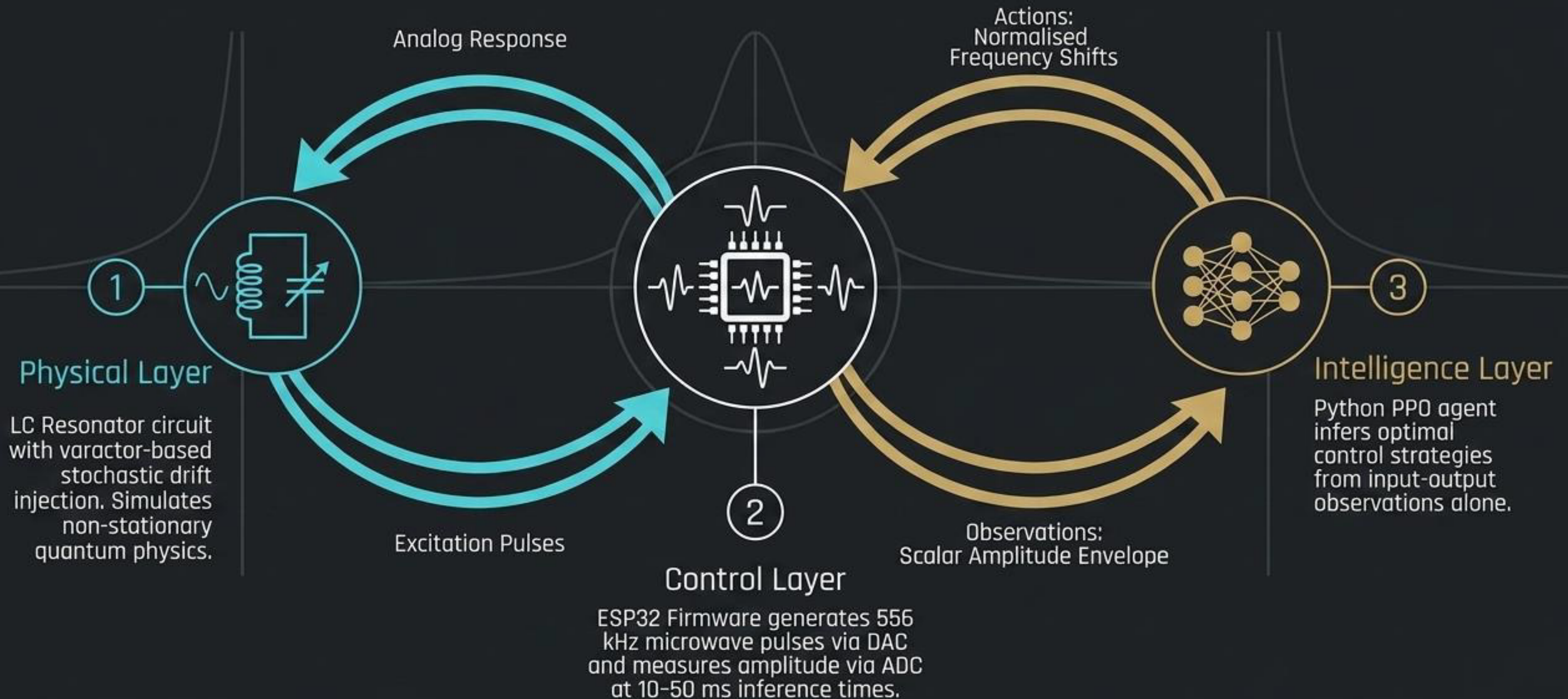
Noise Source:
Voltage-Injected Gaussian Noise
(500 kHz)

The Shared Physics: Both systems generate an identical challenge: tracking a drifting Lorentzian response curve under extreme noise without direct parameter access.

The Core Threat: Gradient Blindness



Methodology I: 3-Layer Closed-Loop Architecture



Methodology II: Engineering the MDP Reward Function

$$\text{Reward} = \text{amp} + 0.5 * (\Delta\text{amp}) * \text{sign}(\text{action}) - 0.01 * |\text{action}|$$

```
graph TD; Eq["Reward = amp + 0.5 * (Δamp) * sign(action) - 0.01 * |action|"] --- PO["Primary Objective"]; Eq --- DB["Directional Bonus"]; Eq --- AP["Action Penalty"];
```

Primary Objective

Maximizes proximity to the resonance peak.

Fails if used alone because it encourages static parking when near the peak.

Directional Bonus

The anti-parking mechanism.

Rewards the agent specifically for taking actions that cause the amplitude to increase, actively forcing it to climb the gradient.

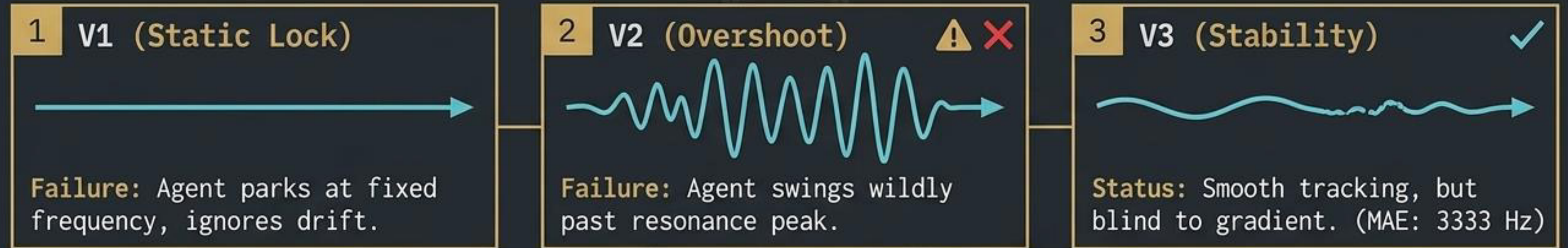
Action Penalty

Prevents erratic, aggressive frequency jumps.

Forces energy and movement efficiency.

Methodology III: The V4 Breakthrough

The Failures



The Fix



Element 5: **amp_gradient**

```
clip(delta_amp * 3000.0 / (|prev_action| + 1.0), -1.0, 1.0)
```

The Impact: Provides explicit Directional Confidence. Normalizes the local slope of the amplitude curve, instantly differentiating self-movement from environmental noise.

Classical Results I: Extreme Gaussian Random Walk

Context: Testing against highly non-stationary stochastic drift ($\sigma=500$ Hz/step).

Diagnostic Comparison Matrix			
	Traditional PID	RL Baseline (V3)	Gradient-Augmented RL (V4)
Mean Absolute Error	3600 Hz (Failure)	3333 Hz	1510 Hz (34% Reduction)
Near-Resonance Time	83.1%	64.5%	94.8%
Standard Deviation (Stability)	$\sigma = 5241$	$\sigma = 1932$	$\sigma = 367$

The 14x variance difference between PID and V4 proves PID suffers from catastrophic episodic failure due to gradient blindness. V4 systematically eliminates this.

Classical Results II: The Dual Noise Proof

Gaussian Drift Regime



V4 completely outperforms V3 ($p < 0.001$).
Unbounded random walk forces deep left/right ambiguity.

Ornstein-Uhlenbeck Drift Regime



V3 and V4 tied ($p = 1.000$). Mean-reverting
drift naturally limits detuning excursions.

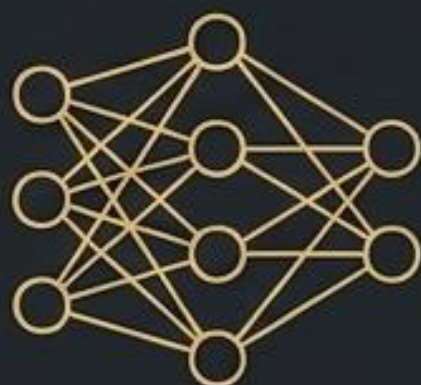
Insight Panel

The Null Result is the Proof. V4 only outperforms V3 when the environment forces extreme directional ambiguity. This proves `amp_gradient` acts as a precise structural fix for `gradient blindness`, not just a generic tuning advantage.

The Quantum Transition: QACT Pipeline

Origin: V4 Seed 3 Weights

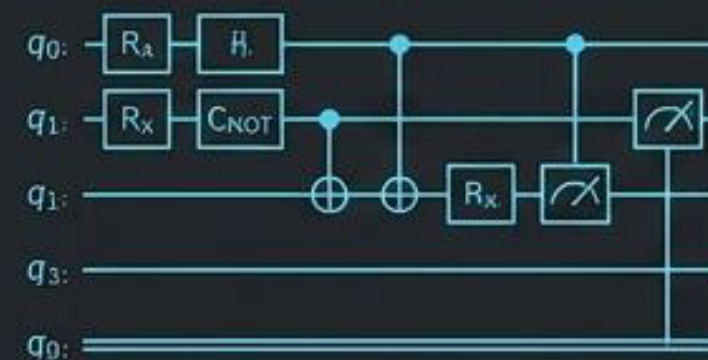
PPO policy trained strictly on Gaussian noise and a classical LC varactor



Zero-Shot
Sim-to-Real Transfer

Destination: Qiskit Aer Simulator

Two-Level System (TLS) quantum circuit



The Physics Translation Matrix

Classical
Amplitude
 $A(f)$

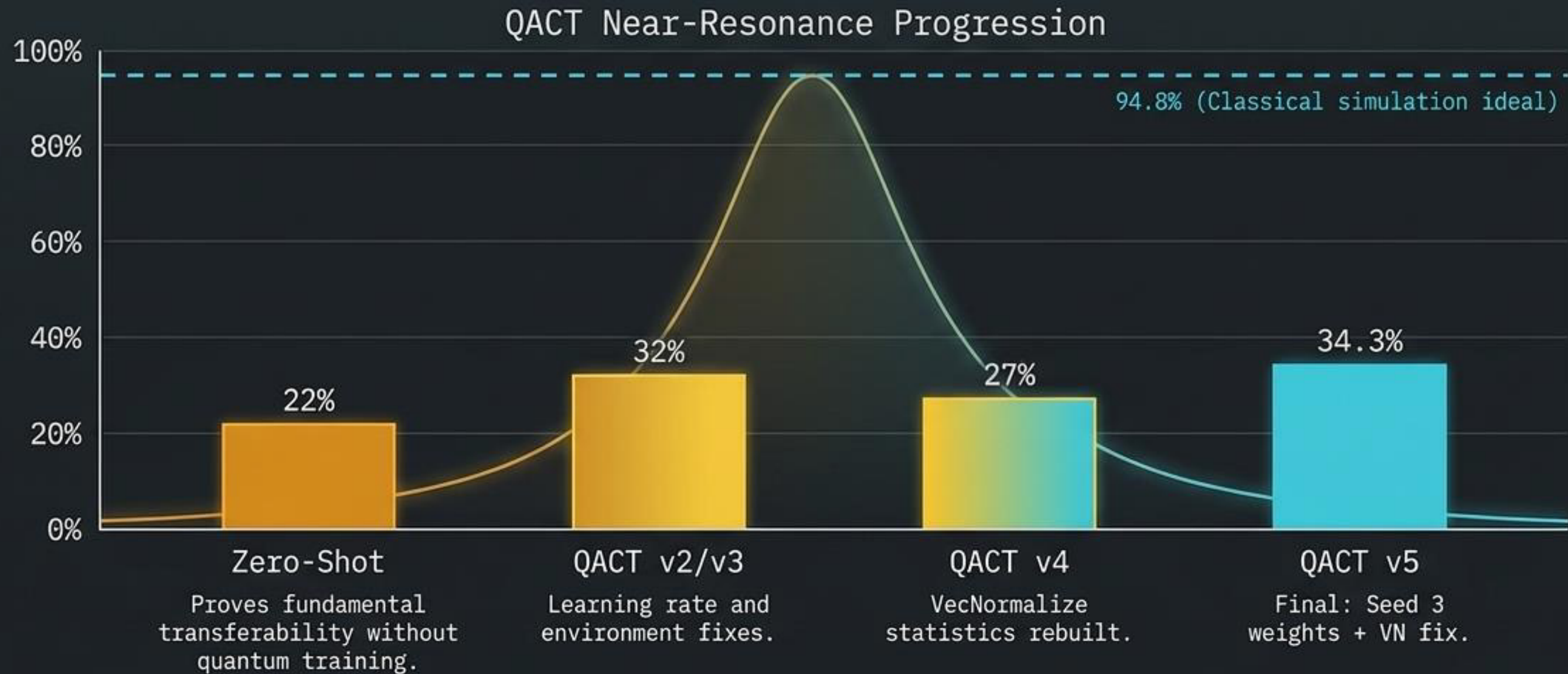


Both share identical Lorentzian geometry.
If the classical agent learned the geometry,
it should control the quantum circuit.

Quantum Excitation
Probability
 $P(|1\rangle)$



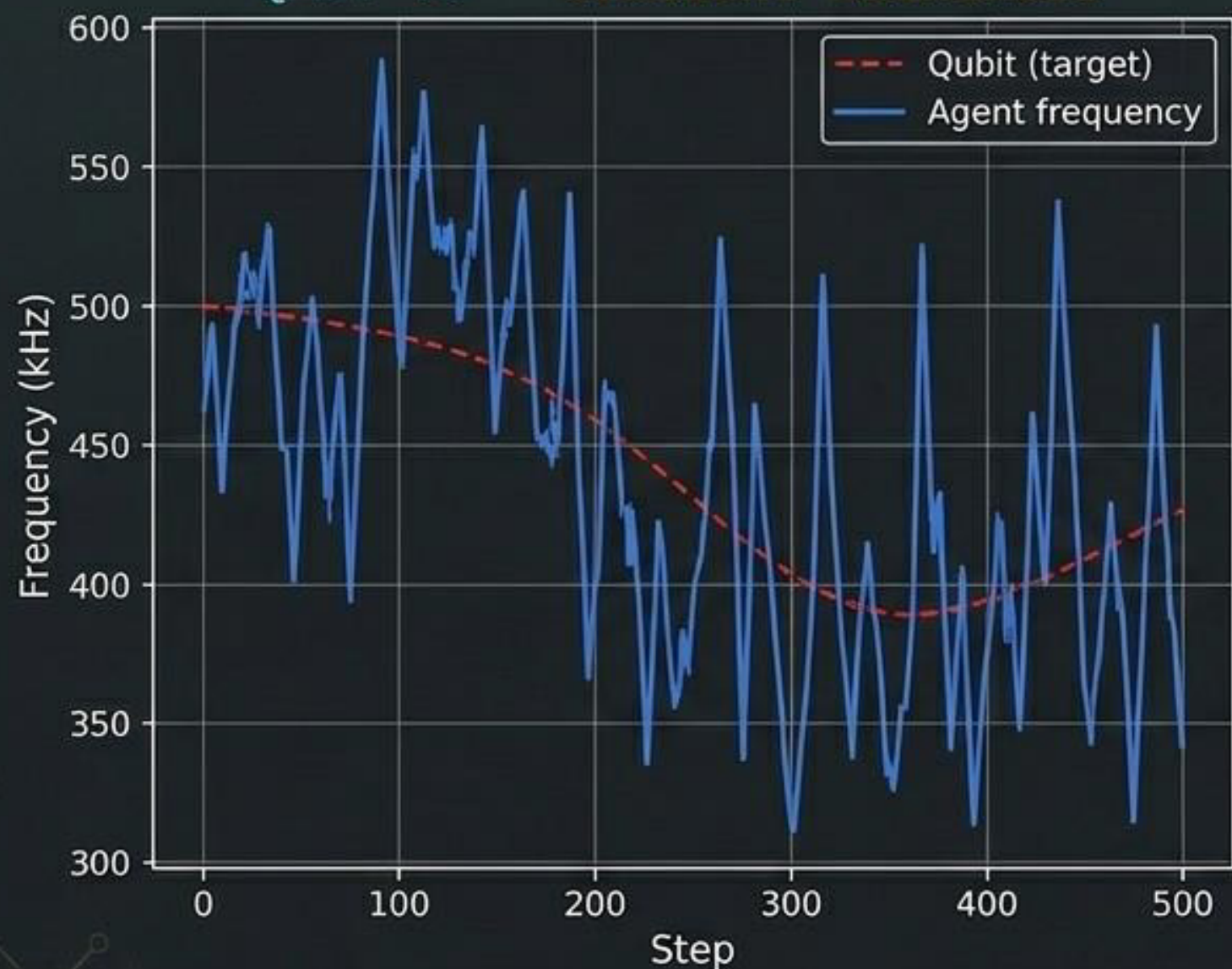
Quantum Results I: Zero-Shot to Fine-Tuned



The Lorentzian hypothesis is validated. A classically trained agent successfully maintains above-random resonance tracking on a quantum circuit.

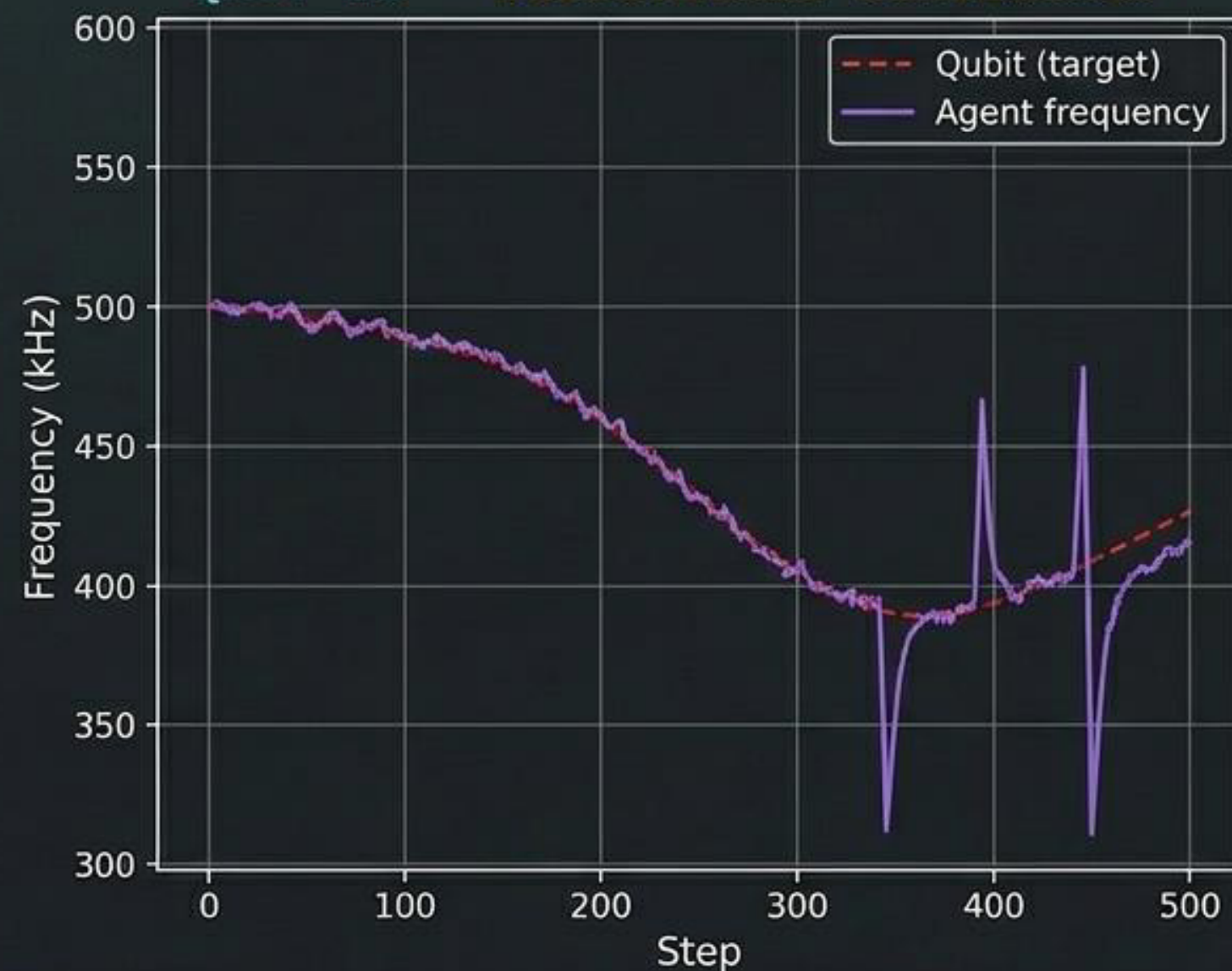
Quantum Results II: Trajectory Analysis

QACT v3 - Gradient Blindness



Agent suffers overshoot. Shot noise destroys unscaled amplitude signals, leaving the agent directionally confused.

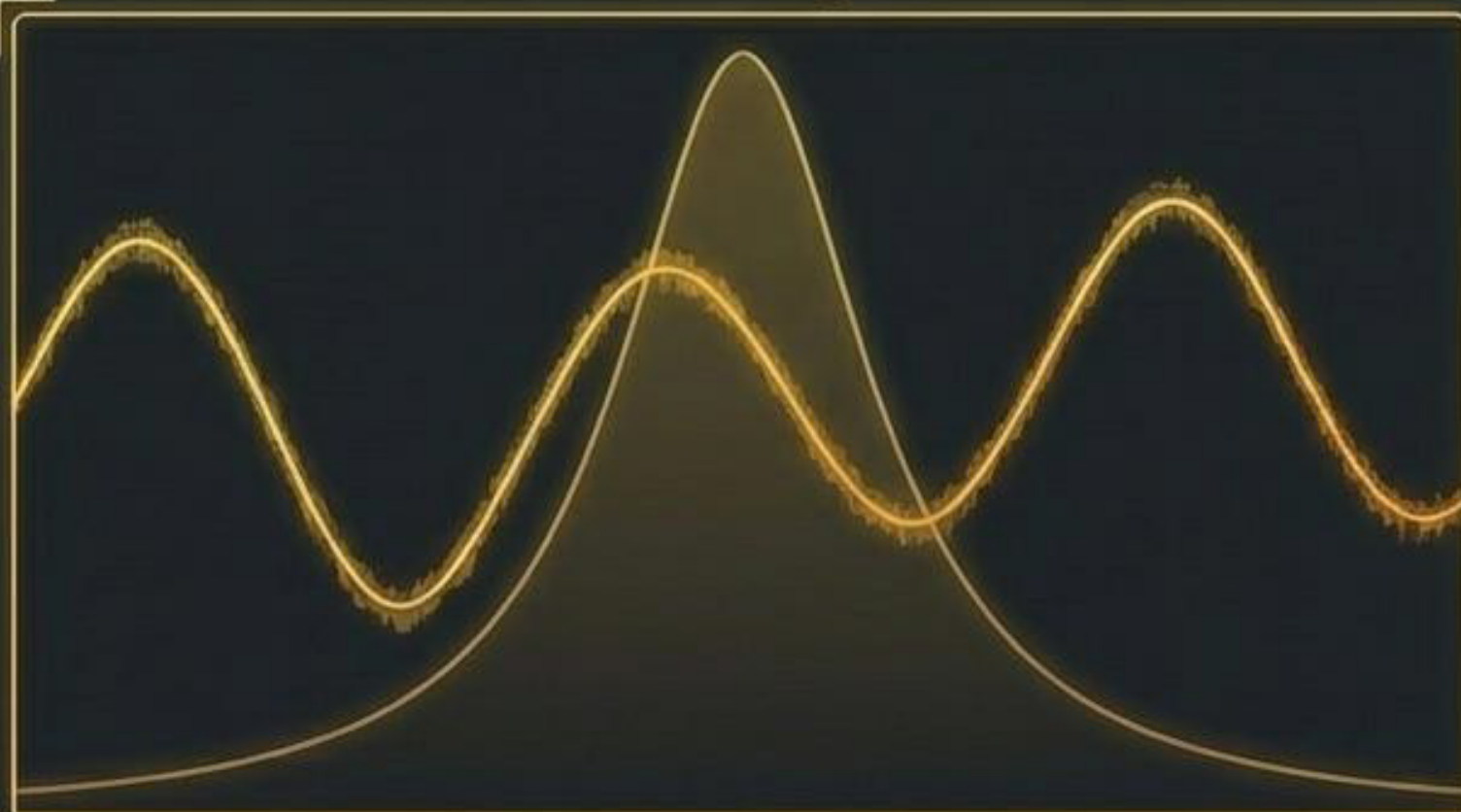
QACT v5 - Directional Confidence



The V4 amp_gradient provides sufficient directional confidence to initiate aggressive, correct recovery sweeps even under quantum measurement noise.

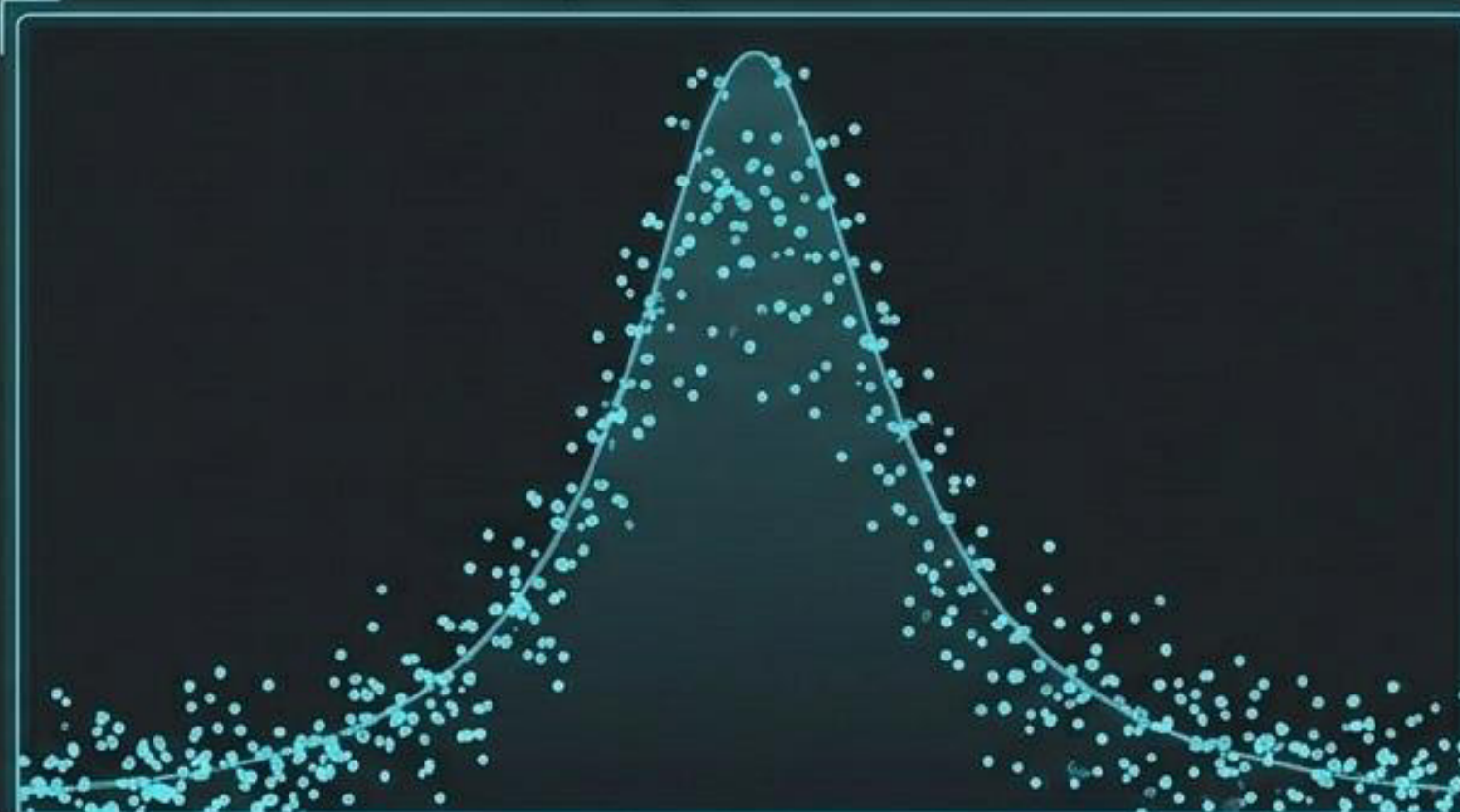
The Transfer Gap: Understanding Quantum Shot Noise

Classical Training Environment



Additive Gaussian Noise.
Deterministic gradient signal.
High SNR allows the agent to read micro-slopes
on the Lorentzian curve perfectly.

Quantum Deployment Environment



Binomial Shot Noise.
Probabilistic measurement.
Shot noise creates false local gradients,
temporarily blinding the amp_gradient feature.

The 60-point drop in near-resonance tracking is a predictable consequence of
shot-noise-limited SNR, dictating the necessity for the final deployment phase.

Implementation: The Hardware Deployment Roadmap



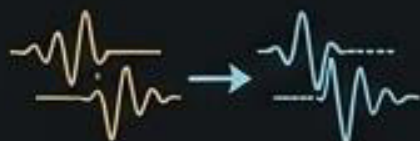
Build 5-stage analog ESP32/DAC/ADC signal chain.



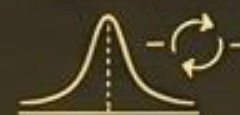
Extract true f_0 , Q-factor, and physical drift rates.



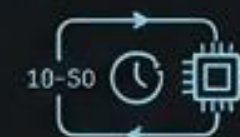
Map the empirical sim-to-real gap.



Adapt policy to
Adapt policy to
real-world hardware
asymmetry and mixed
noise statistics.



Continuous
10-50 ms
inference
deployment.



Hardware-in-the-Loop (HIL) fine-tuning is the explicit architectural fix for the Quantum Shot Noise transfer gap.



Synthesis: The Universal Calibration Analogue



1

Mathematical Rigor

Gradient augmentation structurally solves Left/Right directional ambiguity.

2

Empirical Validation

34% reduction in classical tracking error with a 14x reduction in standard deviation.

3

Quantum Transfer

Validated Zero-Shot and Fine-Tuned sim-to-quantum transfer via QACT pipeline.

By augmenting the observation space with explicit gradient information, this architecture serves as a highly resilient, low-cost classical analogue for next-generation autonomous quantum hardware calibration.